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Causal deep learning for explainable vision-based quality inspection under visual interference

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Abstract

Vision-based quality inspection is a key step to ensure the quality control of complex industrial products. However, accurate defect recognition for complex products with information-rich, structure-irregular and significantly different patterns is still a tough problem, since it causes the strong visual interference. This paper proposes a causal deep learning method (CDLM) to tackle the explainable vision-based quality inspection under visual interference. First, a structural causal model for defect recognition of complex industrial products is constructed and a causal intervention strategy to overcome the background interference is generated. Second, a defect-guided recognition neural network (DGRNN) is constructed, which can realize accurate defect recognition under the training of CDLM via feature-wise causal intervention using two sub-networks with feature difference mechanism. Finally, the causality between defect features and defective product labels can guide the DGRNN to complete the accurate and explainable learning of defect in a causal direction of optimization. Quantitative experiments show that the proposed method achieves recognition accuracy of 94.09% and 93.95% on two fabric datasets respectively, which outperforms the cutting-edge inspection models. Besides, Grad-CAM visualization experiments show that the proposed method successfully captures the data causality and realizes the explainable defect recognition.

Keywords Computer vision · Defect inspection · Deep learning · Causal inference · Explainable artificial intelligence

Abbreviations

CDLM	Causal deep learning method
IQI	Intelligent quality inspection
DL	Deep learning
DNN	Deep neural networks
CIP	Complex industrial products
CPF	Complex patterned fabrics
SCM	Structural causal model
DGRNN	Defect-guided recognition neural network
CIM	Causal intervention module
FDM	Feature difference module
BK-Net	Background knowledge network
DD-Net	Defect detection network
VQA	Visual question and answering
Init_DS	Initial down-sampling
FE	Feature extraction
Conv	Convolution
BN	Batch normalization
FC	Fully-connected
SA	Spatial attention
CA	Channel attention
ACC	Accuracy
PRE	Precision
REC	Recall
AUC	Area under curve

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ROC	Receiver operating characteristic
PAR	Parameter amount
FLOPs	Floating-point operations
FPS	Frame per second

Introduction

Currently, data-driven IQI for industrial products has been widely adopted in production manufacturing and quality control scenarios (Tang et al., 2022; Wang et al., 2022a, 2022b), such as additive manufacturing (Manivannan, 2022), metal products (Zhang et al., 2023), electronic component (Kang et al., 2022) and welding (Liu et al., 2022a, 2022b). Data-driven IQI methods are mainly based on DL (Gao et al., 2021; Xames et al., 2022), which attempts to utilize the feature extraction and learning ability of DNN to establish the data-label correlation to realize the automatic decision-making of the category of input images. However, data-driven IQI methods usually fail when dealing with the defect recognition of CIP, mainly because of the strong visual interferences caused by the manufacturing environment or structural information of product surface. Because the surface of CPF is usually covered with irregular patterns with rich information, it can be regarded as a typical kind of CIP in this scenario. Due to the cumbersome manufacturing process of CPF, defective products commonly occur in the manufacturing links resulting in a significant drop in prices and affecting the economic benefits of textile enterprises (Kumar, 2008). In this paper, CPF are selected as the research objects and a general IQI method for defect recognition of CIP are systematically studied.

The visual interference of CIP can be divided into three types, including graphic inherency interference, graphic similarity interference and graphic difference interference. CPF are taken as an example and three types of the visual interference will be explained as follows in detail.

First, the graphic inherency interference of CPF is caused by their background patterns with rich information and color distribution. As shown in Fig. 1a, the background pattern of simple patterned fabric has regular geometric representations which usually are straight lines or regular curves. Therefore, the amount of information contained in its two-dimensional local entropy map is relatively small. Besides, it can also be seen that the distribution of pixel intensity values of simple patterned fabric is dense, generally within three-unit intervals through the three-channel pixel intensity histogram. However, the geometric representations of CPF are complicated. As shown in Fig. 1b, the background pattern of CPF a is an irregular geometric curve with complex local structure. Therefore, the amount of information represented by its two-dimensional local entropy map is rich and chaotic. In addition, it can be seen from its three-channel pixel intensity histogram that the pixel intensity values of the CPF a are widely distributed, generally within five-unit intervals.

Second, the graphic similarity interference of CPF is caused by the similarity of image information between defect-free and defective CPF. As shown in Fig. 1b, it can be seen that the information distribution of them has a strong similarity. Especially comparing the three-channel pixel intensity distribution histograms of these two images, the difference between the distribution trend of high-intensity and low-intensity pixels and the overall pixel value span is very small, and there are only subtle differences in the distribution of peak and valley pixels distributions.

Third, the graphic difference interference of CPF is caused by the image information difference between various CPF. as shown in Fig. 1c, although the geometric representation of CPF b is also the irregular geometric curves, the information contained in its characteristic structure and distribution is quite different from CPF a. And there are two obvious differences in the three-channel pixel intensity histograms of CPF a and CPF b, which are the high-low intensity pixel distribution trend and peak-valley pixel intensity distribution between the two fabric images. Generally speaking, these three kinds of visual interference can affect the data-driven IQI methods to achieve accurate defect recognition and detection.

The reason why the data-driven IQI methods fail to realize the defect recognition of CIP is that the feature extraction process of DNN is unexplainable, which reflects the limitations of DL-based algorithms applied on the defect recognition of industrial products under visual interference. For supervised image classification task, e.g., defect recognition of CPF, the essence of traditional DL-based algorithms is to use loss functions and gradient descent method to optimize model parameters to minimize empirical risk and learn the feature-label correlation to further detect the anomaly area of the product images. However, the problem is that DNN is unable to distinguish whether the learned correlation is causal, i.e., defect features correspond defective product label, where only causality can drive a model make explainable decisions (Cui & Athey, 2022). In the defect recognition of CIP, causality can be defined as the unbiased correlation between visual features and image labels, which has objective inherent relationship. When DNN processes an image of defective product, it is expected that it can assign the defective product label based on precise recognition and sampling of the defective area, which follows the unbiased causality between defect features and defective product labels. Due to the coexistence of background features and defect features, the DNN may stuck in the pixel-wise sample selection bias and spurious correlation dilemma, i.e., the background features are sampled as defect features and used to establish the correlation between defect features and defective product labels.

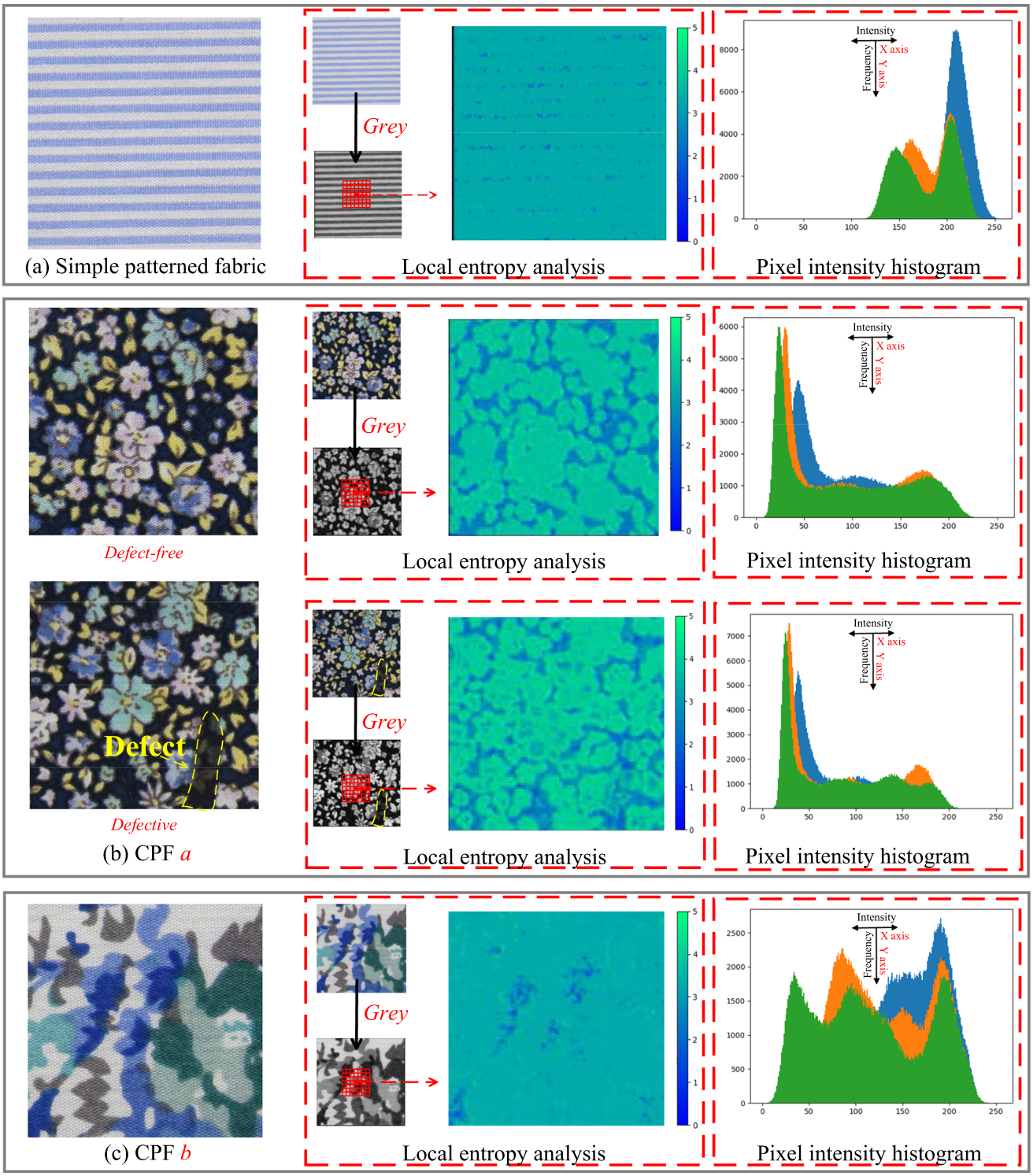


Fig. 1 The comparison between simple and complex patterned fabrics

The above phenomena all point to the challenging issue that the causality contained in the CIP is hard to capture, thus the data-driven IQI methods cannot be directly adopted in the explainable defect recognition of CIP. Consequently, a CDLM for explainable defect recognition of CIP is proposed. First, causal inference (Pearl, 2009; Kuang et al., 2020; Schölkopf & Kügelgen, 2022) is introduced to study the causal effects between all variables in defect recognition of CIP by a SCM, and a causal intervention strategy for explainable and accurate defect recognition is generated. Second, a DGRNN is proposed which contains BK-Net, DD-Net and CIM. CIM completes the feature-wise causal intervention by differencing the output features from the two above-mentioned sub-networks, and solves the graphic inherency interference and similarity interference problems. In addition, a combination strategy of multi-scale FDM and causality expansion factor is proposed to solve the problem of graphic difference interference. Finally, CIM outputs the intervention weights of multi-scale difference calculations to construct a causality guidance loss to optimize DGRNN to achieve accurate extraction and explainable recognition of defects under visual interference.

The rest parts of this paper are arranged as follows. In "Related works" section, literatures of data-driven IQI and causality-based explainable computer vision are reviewed. In "Causal inference for defect recognition of CIP" section, the procedure about causal inference of defect recognition of CIP is introduced. In "Causal deep learning method" section, the proposed CDLM and DGRNN are described in detail. The experimental settings and result analysis are shown in the "Experimental setting" and "Result and analysis" section. Finally, in "Conclusion" section, the conclusions and future works of this paper are presented.

Related works

The related researches can be summarized as the following two aspects: (1) data-driven IQI; (2) causality-based explainable computer vision.

Data-driven IQI

Data-driven IQI is at the forefront of smart manufacturing. According to the characteristics of manufacturing scenarios in various industrial products, IQI methods are also very different. Due to the special structure of the wafer products, there are mixed-type defects presented in the wafer maps. To solve this problem, a neural network with a knowledge augmented module and multi-channel selective extraction is proposed to decouple the multiple defects and achieve visual inspection of mixed-type defects of wafers (Wang et al., 2023). In the laser welding scene, because the visual differences between various types of molten pools/keyhole defects are inconspicuous, an attention-based bilinear feature extraction mechanism is used to achieve fine-grained recognition of molten pools/keyhole defects (Liu et al., 2023). Since it is difficult to obtain labeled data in steel manufacturing, a semi-supervised learning-based inspection network is proposed, which can achieve accurate steel surface defect inspection with requirement of fewer labeled samples (Gao et al., 2020). Due to blurry and vapor interference from the images collected in underground sewer, an instance segmentation method Pipe-SOLO using dehazing generative adversarial network is proposed, which can realize the accurate inspection of six kinds of sewer defects (Li et al., 2022).

Fabric quality inspection has always been a heated topic in manufacturing systems. Early vision-based fabric quality inspection methods focused on the enhancement of texture features (Chan & Pang, 2000; Ngan et al., 2005; Mak et al., 2009), and it is difficult to guarantee the automatic and efficient inspection of defects. In contrast, current IQI technology for fabrics benefits from the excellent feature extraction and learning capabilities of the DNN, and has breakthroughs in lightweight and efficient inspection (Jing et al., 2022; Li et al., 2019; Wang et al., 2022a, 2022b), micro-defect inspection (Wei et al., 2020), and generalizable inspection (Liu et al., 2019).

However, the above-mentioned IQI methods are mostly aimed at the quality inspection of gray fabrics or simple texture fabrics, which do not have strong visual interference like the CIP. At present, there is still a lack of solutions to tackle the quality inspection problem of CIP under visual interference from the perspective of explainability. A small number of researches on the interference of fabric surface textures focus on the use of biological visual attention and visual memory methods to enhance the ability of DNN to extract defect features (Li et al., 2018; Zhao et al., 2020a, 2020b). In welding scenes, an explainable recognition method for laser welding status is proposed (Liu et al., 2022a, 2022b). While the multi-scale quality feature is extracted by the CNN, it is simultaneously input to the class activation map branch network for visualization of the recognition decision. Lee et al. (2022) proposed an explainable analysis method for visual inspection models of the thin-film-transistor liquid–crystal display. First, multi-layer defects are visualized through a layer-wise relevance propagation method and input into a decision tree model for prediction. Finally, the explainability of the model is analyzed by comparing the prediction results with expert prior knowledges and rules. Besides, there are also some studies that directly adopt explainable methods, e.g., Karthikeyan et al. (2022) and Sundarrajan and Rajendran (2023) adopted local interpretable model-agnostic explanations (Ribeiro et al., 2016) to tackle the explainability issues of the defect inspection in ultrasound images and metal surface. However, these methods adopt the post-hoc

analysis methods that cannot assist the model to realize the explainable learning and optimization, which indicates the inherent shortage of data-driven IQI methods that do not run in a causal way remains unaddressed.

Causality-based explainable computer vision

Causality is a reliable statistical correlation among data, which enables data-driven vision models to make explainable decisions (Cui & Athey, 2022). In recent years, researches combining causal inference with computer vision have begun to emerge (Cui et al., 2020; Schölkopf et al., 2021). Since images are unstructured data so that it cannot directly provide causal information (Schölkopf et al., 2021), earlier works focused on VQA tasks (Abbasnejad et al., 2020; Wang et al., 2020; Yang et al., 2021). Wang et al. (2020) proposed a novel R-CNN framework to realize accurate VQA task, which utilizes a dictionary containing all potential confounding factors to simulate backdoor adjustment and achieve the intervention of confounding features and accurate question answering. Yang et al. (2021) proposed a causal language attention mechanism CATT is proposed to eliminate potential confounding effects in visual models by simulating causal intervention. Abbasnejad et al. (2020) proposed a visual language learning method incorporating counterfactual inference is proposed to increase the generalization ability of VQA models. In addition to solve the VQA tasks, causality-based computer vision has also been extended to segmentation tasks. A causality-driven robotic tool segmentation method is proposed to achieve accurate robotic tool image segmentation by considering the causal relation between variables such as system and surgical environment and segmented images (Ding et al., 2022). A context adjustment method CONTA for weakly supervised semantic segmentation is proposed, which implement causal intervention on context mixing factors to eliminate its confounding influences (Zhang et al., 2020a, 2020b).

In manufacturing systems, the application of causality in industrial data analysis has been developed (Sun et al., 2021, 2022; Zhou et al., 2021). However, there is still a lack of research integrating causality in the field of industrial image processing, especially in the field of defect recognition of CIP. The current computer vision method based on causality can mainly be divided into the following three steps. First, to construct a SCM for a particular task. Second, to analyze the causal intervention strategy through causal inference. Finally, to design a DNN-based model or learning strategy to implement the causal intervention strategy. However, for defect recognition of CIP, there are still two things to be considered. First, how to perform causal inference on the downstream task of CIP defect recognition and get the causal intervention strategy. Second, how to implement causal intervention strategy using DNN or DL methods. Therefore, the above-mentioned research works and gaps motivate us to combine causal inference and the industrial image processing tasks, and apply them to defect recognition of CIP.

Causal inference for defect recognition of CIP

In this section, the causal inference of accurate defect recognition of CIP will be introduced in detail, mainly consisting of the following two parts. First, a SCM is constructed for the defect recognition of CIP. Second, a causal intervention strategy that can block the interference of confounding features is obtained through backdoor adjustment.

Construction of SCM for defect recognition of CIP

As shown in Fig. 2a, the defect recognition task of CIP is represented by four variables, which are pixels, defect features, background features, and defective product labels. Therefore, the defect recognition of CIP can be constructed as a SCM shown in Fig. 2b, where node U represents the pixels, node X represents the defect features, node Z represents the background features and the node Y represents the defective product labels.

a. $U \rightarrow X \rightarrow Y$: This part indicates that the DNN samples the defect features from the pixel matrix and uses them to infer the defective product label, which can be called the causal path and its observed variables and causal functions are shown in Eqs. (1) and (2). Causal path is the ideal path to achieve accurate and explainable defect recognition, which contains the causality between defect features and defective product labels.

$$n = \{U,\ X,\ Y\} \tag{1}$$

$$f_c = \left\{f_{ux} : U \rightarrow X,\ f_{xy} : X \rightarrow Y\right\} \tag{2}$$

b. $U \rightarrow Z \rightarrow Y$: This part indicates that the DNN samples the background features from the pixel matrix and uses them to infer the defective product label, which can be called the confounding path and its observed variables and causal functions are shown in Eqs. (3) and (4). The spurious correlation between background features and defective product labels is included in the confounding path, which affects explainable and accurate defect recognition.

$$n = \{U,\ Z,\ Y\} \tag{3}$$

$$f_c = \left\{f_{uz} : U \rightarrow Z,\ f_{zy} : Z \rightarrow Y\right\} \tag{4}$$

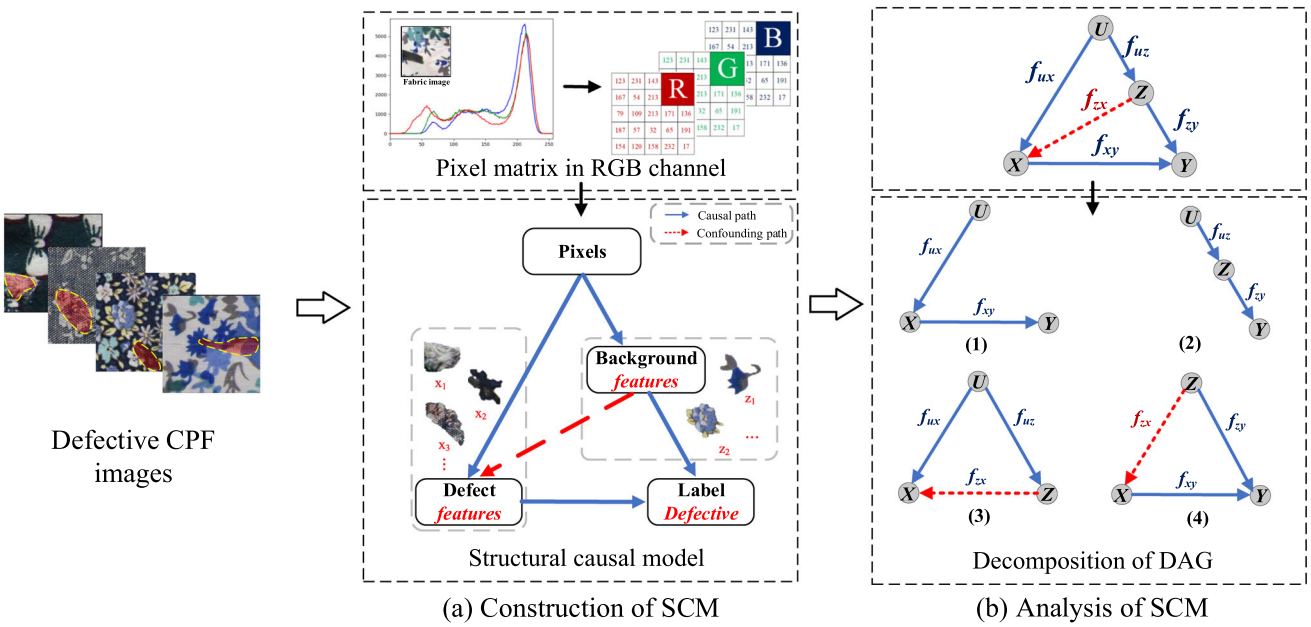


Fig. 2 The causal inference of defect recognition of CIP

c. $X \leftarrow U \rightarrow Z$: This part indicates that the pixel matrix contains pixel regions representing defect features and background features, which describes the relationship between defect features and background features, and its observed variables and causal function are shown in Eqs. (5) and (6). According to the causal inference, node U is the parent node of node X and node Z, which means this the pixel is the common cause of the defect features and the background feature resulting in a fork structure formed by these three nodes. The fork structure leads to an uncertain correlation between node X and node$Z$.

$$n = \{U,\ X,\ Z\} \tag{5}$$

$$f_c = \{f_{ux} : U \rightarrow Y,\ f_{uz} : U \rightarrow Z,\ f_{xz} : X \rightarrow Z,\ f_{zx} : Z \rightarrow X\} \tag{6}$$

d. $Y \leftarrow Z \leftrightarrow X \rightarrow Y$: This part indicates that the background features have confounding effects on the defect features, and its observed variables and causal function are shown in Eqs. (7) and (8). This path can be interpreted as the background features have an observable interference on the causal path of defect recognition, and the model cannot learn the features that have a causality with the defective product label.

$$n = \{Z,\ X,\ Y\} \tag{7}$$

$$f_c = \left\{f_{zy} : Z \rightarrow Y,\ f_{xy} : X \rightarrow Y,\ f_{zx} : Z \rightarrow X,\ f_{xz} : X \rightarrow Z\right\} \tag{8}$$

In general terms, the DNN-based inspection model simultaneously establishes the common correlation of defect features, background features and defective product labels when down-samples the high-dimensional pixel matrix of defective CIP images. Therefore, the background features interfere with the decision-making of the model using the causality between defects features and defective product labels as confounding features.

Causal intervention strategy for defect recognition of CIP

In order to achieve accurate and explainable defect recognition of CIP, this section will calculate a causal intervention strategy to block the confounding path. By analyzing the SCM, it can be seen that the ideal distribution between features and labels should satisfy the Bayesian formula $P(Y|X)$. Under this ideal distribution, the data correlation between features and labels learned by the DNN-based model must be the expected causality for the defect recognition task. However, due to the existence of background, the actual distribution satisfies $P(Y|X,\ Z)$, which defect features and background features co-exist in the feature space. Therefore, the do-calculus is introduced into the Bayesian formula of the ideal distribution shown in Eq. (9). To eliminate uncalculatable do, backdoor adjustment is used to obtain the causal intervention strategy shown in Eq. (10).

$$P(Y|do(X)) = \sum\nolimits_Z P(Y|do(X),\ Z)P(Z|do(X)) \tag{9}$$

$$P(Y|do(X)) = \sum\nolimits_Z P(Y|X,\ Z)P(Z) \tag{10}$$

Therefore, the ideal distribution for accurate defect recognition of CIP can be approximated using causal intervention strategies shown in Eq. (10). In the next section, the implementation method of the causal intervention strategy will be described in detail.

Causal deep learning method

In this section, to implement the proposed causal intervention strategy for defect recognition of CIP, the CDLM is proposed in this section. First, the DGRNN will be constructed. Second, the operation of CDLM is introduced.

Construction of DGRNN

The DGRNN consists of two sub-networks with a feature difference mechanism, namely the DD-Net and the BK-Net, and the structure of DGRNN is shown in Fig. 3.

DD-Net is designed to achieve accurate extraction and recognition of CIP defect and its operation process represents $P(Y|X, Z)$ in the causal intervention strategy. BK-Net is designed to learn and extract the background features from CIP with multiple backgrounds and its operation process represents $P(Z)$ in the causal intervention strategy. BK-Net can also be seen as a prior knowledge set of confounding factors containing background features of CIP. In addition, CIM for implementing causal intervention strategy is proposed, which includes four FDM to realize feature difference in multiple-scales.

Structure of DD-Net

First, the structure of DD-Net is described. As shown in Fig. 3, DD-Net is a defect recognition network that can make decision of the class of input CIP images. DD-Net includes five components, which are Init_DS_layer, FE_layer, Self-Attention layer, adaptive average pooling layer (Adaptive_avgpool_layer) and FC_layer. The Init_DS_layer consists of two Conv layers and BN layers, which is designed for down-sampling the input CIP image to half the original size and expanding its channels to 64. Its parameters of each layer are shown in Table 1.

The FE_layer is designed for down-sampling the feature maps output by the Init_DS_layer to half of the original size while keeping the channels unchanged. ReLu is adopted to project the output feature maps into the non-linear space. Its parameters of each layer are shown in Table 2.

The second, third and fourth FE_layers are designed for down-sampling the feature maps output from the previous layers to half of the original size and double the channels at the same time. The parameters of each layer are shown in Table 3.

At the end of the DD-Net, an Adaptive_avgpool_layer is used to down sample the input feature map. Finally, features are flattened and input to the FC_layer for classification. Besides, in order to improve the network sensitivity to extract the key features, the Self-Attention_layer (Woo et al., 2018) is added at the end of each FE_layer. As shown in Fig. 4, the Self-Attention_layer contains CA block and SA block, which obtain the channel-wise and position-wise attention weights and generate attention feature maps respectively. Specifically, the CA block processes the feature map of each channel by using max-pooling and average-pooling respectively, and feeds the output vector into a multi-layer perceptron with three linear layers. Then, the output vectors are added by elements and activated to obtain CA weights which are further weighted to the original feature maps. The SA block processes the feature map at the same position by using max-pooling and average-pooling respectively, and feeds the output vector into a Conv layer. The output vectors are activated to obtain SA weights which are further weighted to the original feature maps.

Structure of BK-Net

Second, the structure of BK-Net is described. As shown in Fig. 3, the overall network structure of BK-Net is basically the same as that of DD-Net. The main difference is that in order to achieve the end-to-end output of the background features, the Adaptive_avgpool_layer and the FC_layer at the end of the network are both removed after convergence.

Structure of CIM

In order to achieve precise learning of the causality between defect features and defective product labels, CIM is proposed. By analyzing the learning process of BK-Net and DD-Net, CIM constructs a causality guidance loss to realize the explainable optimization of the DGRNN in a causal direction. As shown in Fig. 3, CIM contains four FDM, which are set in the four main down-sampling stages of DGRNN respectively. As shown in Eq. (11), each FDM differences the feature maps output from the corresponding scales of BK-Net and DD-Net to obtain the feature difference representation S_i. Then, S_i is input into a built-in Adaptive_avgpool_layer to reduce the dimension which can also keep the dimensions consistency of S_i at each scale. Subsequently, a channel-wise average of S_i is calculated and the flattened S_i is projected to a non-linear space by sigmoid function to obtain the final feature difference weight ω_i at a one scale, which is shown in Eq. (12). In order to balance the impact of the four scale difference weights on feature recognition, a causal expansion factor λ_i is introduced to weight and sum the multi-scale difference weights to obtain the final causal intervention weight ω_f which is shown in Eq. (13). Finally, ω_f is used to construct

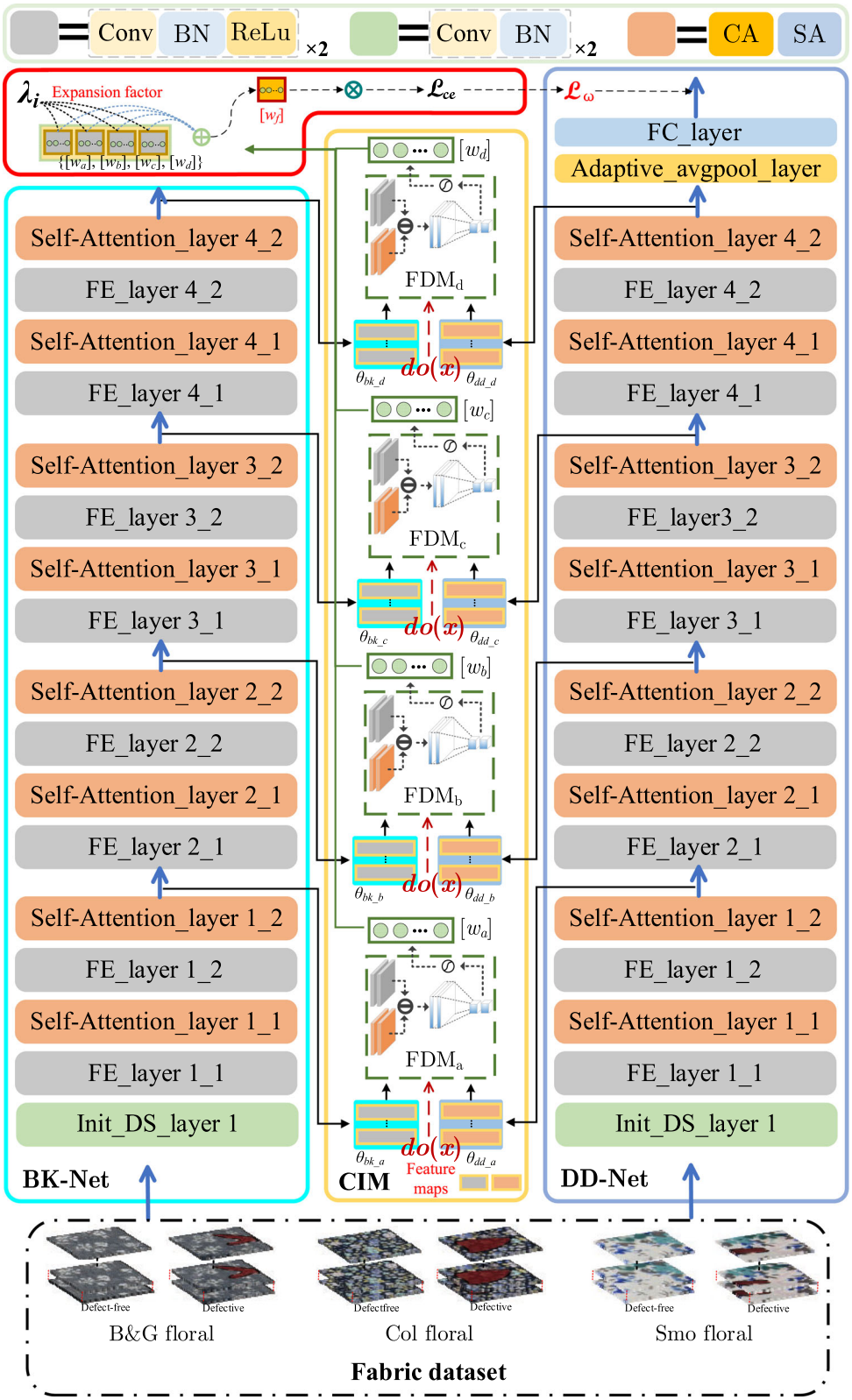


Fig. 3 The structure of DGRNN

Table 1 The structure of Init_DS_layer

Layer	Kernel size	Stride	Padding	Input size	Output size
Conv 1	7 × 7	2	3	[n,3,224,224]	[n,32,112,112]
BN	–	–	–	[n,32,112,112]	[n,32,112,112]
Conv 2	1 × 1	1	0	[n,32,112,112]	[n,64,112,112]
BN	–	–	–	[n,64,112,112]	[n,64,112,112]

Table 2 The structure of FE_layer 1

Layer	Kernel size	Stride	Padding	Input size	Output size
Conv 1	3 × 3	2	1	[n,64,112,112]	[n,64,56,56]
BN	–	–	–	[n,64,56,56]	[n,64,56,56]
ReLu	–	–	–	[n,64,56,56]	[n,64,56,56]
Conv 2	3 × 3	1	1	[n,64,56,56]	[n,64,56,56]
BN	–	–	–	[n,64,56,56]	[n,64,56,56]
ReLu	–	–	–	[n,64,56,56]	[n,64,56,56]

Table 3 The structure of FE_layer 2, 3, 4

Layer	Kernel size	Stride	Padding	Input size	Output size
Conv1	3 × 3	2	1	[n,c,h,w]	[n,2c,h/2,w/2]
BN	–	–	–	[n,2c,h/2,w/2]	[n,2c,h/2,w/2]
ReLu	–	–	–	[n,2c,h/2,w/2]	[n,2c,h/2,w/2]
Conv2	3 × 3	1	1	[n,2c,h/2,w/2]	[n,2c,h/2,w/2]
BN	–	–	–	[n,2c,h/2,w/2]	[n,2c,h/2,w/2]
ReLu	–	–	–	[n,2c,h/2,w/2]	[n,2c,h/2,w/2]

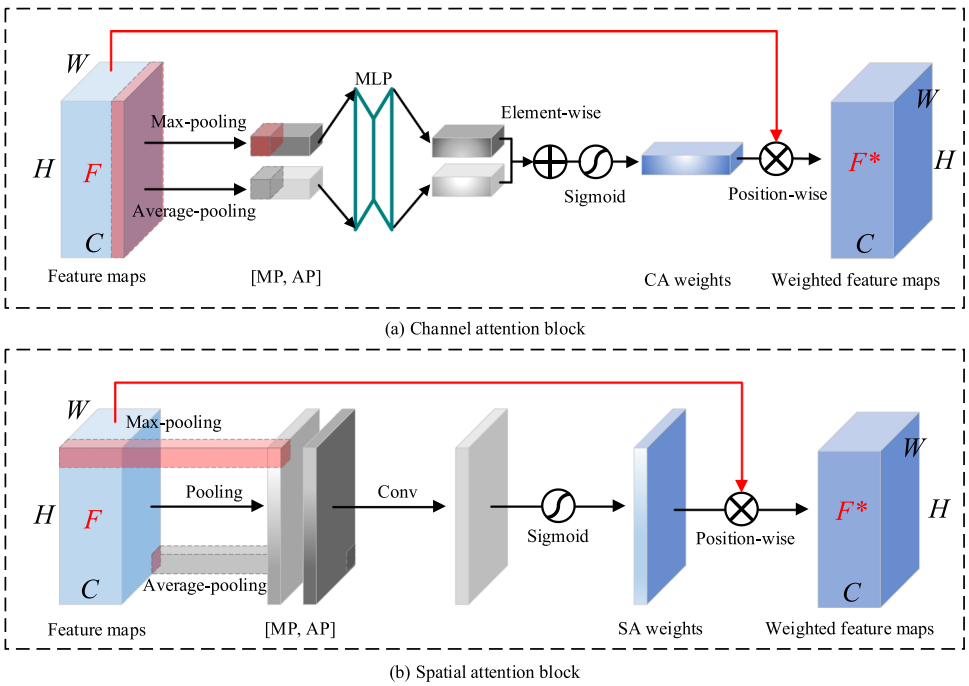


Fig. 4 Self-Attention layer

causality guidance loss which guides DD-Net to only learn defect features and avoid background features interference. The causality guidance loss is shown in Eq. (14).

$$S_i = F(f_{ext,\theta_i^t}(x_i^t) - f_{ext,\theta^p}(x_i^t)) \tag{11}$$

where x_i^t denotes the i-th input image, $F(\cdot)$ denotes the adaptive_avgpool operation, $f(\cdot)$ denotes the feature extractor of DGRNN at on scale, and θ^t, θ^p denote the parameters of DD-Net and BK-Net respectively.

$$\omega_i = \frac{1}{1+e^{-S_i}} \tag{12}$$

$$\omega_f = \sum_{i=1}^{4} \lambda_i \omega_i \tag{13}$$

$$L_\omega = -\sum_{i=1}^{b} \omega_f (y_i^t \cdot \log(f_{cls}(f_{ext}(x_i^t)))) \tag{14}$$

where $f_{ext}(\cdot)$ and $f_{cls}(\cdot)$ denote the feature extractor and classifier of DD-Net.

The operation of CDLM

The CDLM can be divided into two phases which are the pre-training of BK-Net and causality-guided training of DD-Net. The pseudocode of the CDLM is shown in Fig. 5.

Algorithm 1: CDLM

Input: $\mathcal{D}_m^{train} = \{(x_1^t, y_1^t), ..., (x_n^t, y_n^t)\}$;
$\mathcal{D}_m^{pre-train} = \{(x_1^p, y_1^p), ..., (x_n^p, y_n^p)\}$; $batch\ size = b$;

Ouput: Converged DGRNN with θ^p, θ^t

Phase I: <Pre-training of BK-Net>

foreach $k \in epoch$ **do**
  foreach $Minibatch\ X = \{x^p, y^p\}_{j=1}^{b} \in \mathcal{D}_m^{pre-train}$ **do**
    Minimize $\mathcal{L}_{CE} = -\sum_{i=1}^{b} y_i^p \log(f_{cls}(f_{ext}(x_i^p)))$;
    Update θ_k^p ;
  end
end

Phase II: <Causality-guided training of DD-Net>

Initialize $\theta_i^t = \theta^p$;
foreach $k \in epoch$ **do**
  foreach $Minibatch\ X = \{x^t, y^t\}_{j=1}^{b} \in \mathcal{D}_m^{train}$ **do**
    Calculate $S_i = F(f_{ext,\theta_i^t}(x_i^t) - f_{ext,\theta^p}(x_i^t))$;
    Calculate $\omega_i = \frac{1}{1+e^{-S_i}}$;
    Calculate $\omega_f = \sum_{i=1}^{4} \lambda_i \omega_i$;
    Minimize $\mathcal{L}_\omega = -\sum_{i=1}^{b} \omega_f (y_i^t \cdot \log(f_{cls}(f_{ext}(x_i^t))))$;
    Update θ_k^t ;
  end
end

Fig. 5 The pseudocode of CDLM

In phase one, BK-Net is trained to distinguish the background feature representation of multiple defect-free CIP. The pre-training dataset $D_m^{pre-train}$ only contains multiple defect-free CIP images. The specific process is as follows. First, to initialize the parameters θ^p of BK-Net. Second, to randomly sample a batch of CIP images from $D_m^{pre-train}$ and input it into BK-Net for classification. Then, to update θ^p by minimizing the cross-entropy loss until the model converges, where the cross-entropy loss is shown in Eq. (15). Finally, the final parameters of BK-Net are saved at the end of training. Through the first stage of training, BK-Net has the prior knowledge about background features of CIP.

$$L_{CE} = -\sum_{i=1}^{b} y_i^p \log(f_{cls}(f_{ext}(x_i^p))) \tag{15}$$

In phase two, the causal intervention of background features is realized through CIM, and the causality guidance loss is constructed by using the intervention weight ω_f to enable DD-Net learn the causality between defect features and defective product labels. The dataset D_m^{train} contains defect-free and defective CIP with multiple backgrounds. The specific process is as follows. First, the final parameters θ^p of BK-Net are transferred to DD-Net as its initial parameters θ^t to ensure the consistency of the feature representation. Second, to randomly sample a batch of CIP images from D_m^{train} and input it into the converged BK-Net and the DD-Net to be trained at the same time, and the distribution of background features is intervened by calculating the feature difference representation from the two sub-nets. At this stage, parameters of BK-Net are not updated and BK-Net only performs end-to-end feature extraction. Then the feature difference weight ω_i of four scales are calculated separately, and the causal expansion factorλ_i is used to calculate the intervention weight ω_f in the current batch. Finally, causality guidance loss is constructed using ω_f, while minimizing this loss function and updating parameters until the DD-Net converges. Finally, the final parameters of DD-Net are saved. Through the second stage of training, DD-Net is able to achieve accurate and explainable defect recognition of CIP.

Experimental setting

The proposed method is built on Ubuntu 18.04.6 system, compiled under the framework of Python 3.9 and Pytorch 1.11, and executed on CPU x Intel (R) Xeon (R) CPU E5-2698 v4 @ 2.20 GHz and NVIDIA Tesla V100 32 GB. The datasets selected in this paper are based on ZJU-Leaper (Zhang et al., 2020a, 2020b). ZJU-Leaper contains 5 groups of 19 kinds of colored fabric images and the reason why it is

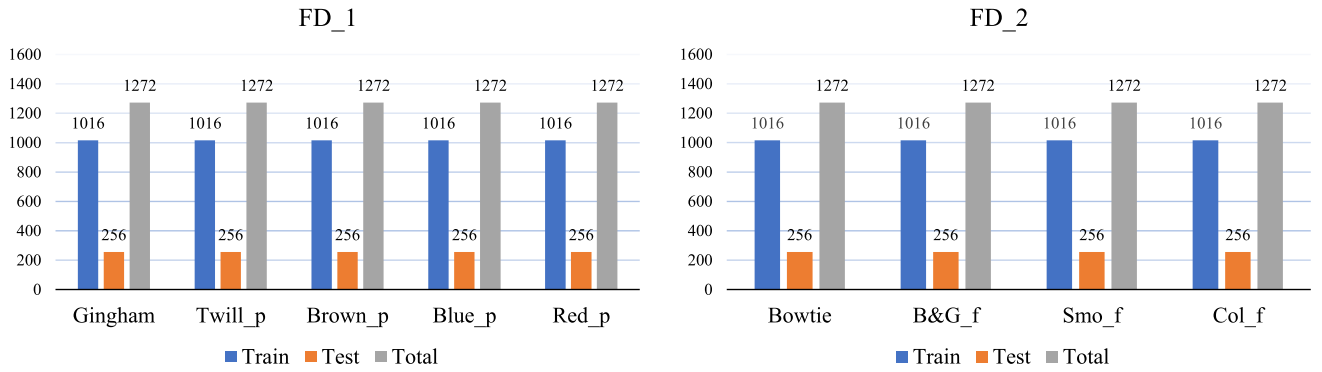


Fig. 6 The statistics of $D_m^{pre-train}$

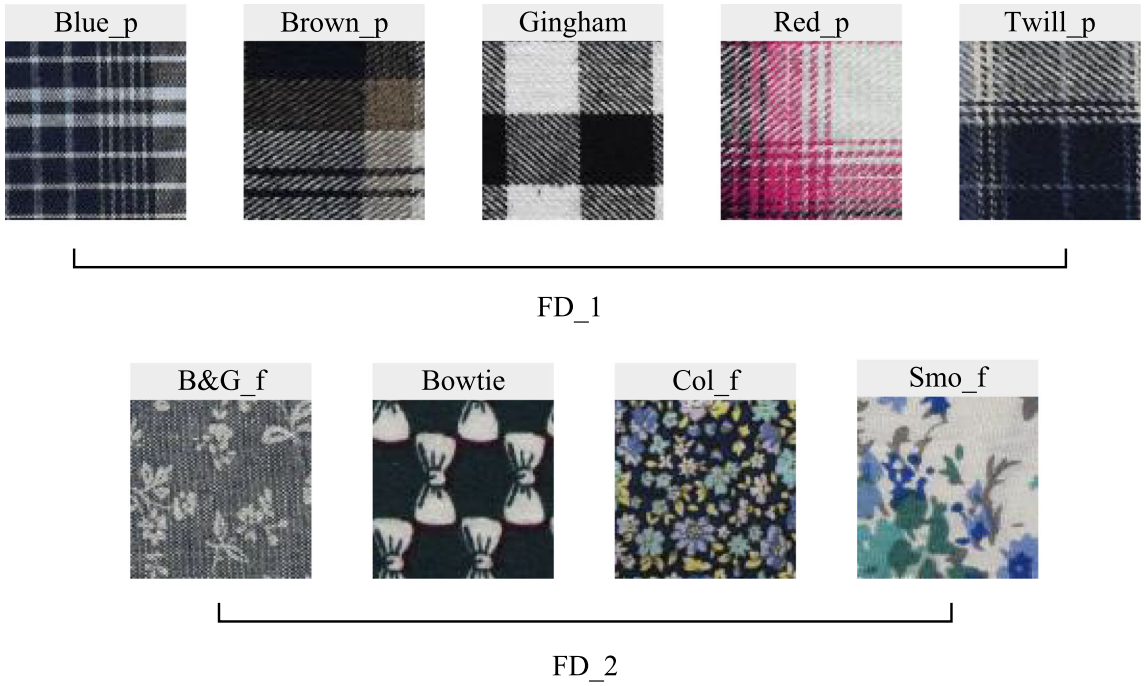


Fig. 7 The samples of $D_m^{pre-train}$

superior to other fabric datasets is that it contains a variety of fabric images with irregular and complex patterns, which can be extended to a variety of complex defect recognition researches. The data used in this experiment include the CPF images sampled from ZJU-Leaper and are divided into two datasets, namely the pre-training $D_m^{pre-train}$ and the defect inspection training dataset D_m^{train}, the statistics and samples are shown in Figs. 6, 7, 8 and 9. All datasets contain two sub-datasets of CPF, namely FD_1 whose fabrics are characterized by regular geometric straight lines, and FD_2 whose fabric is characterized by irregular geometric curves.

Due to the limited data size of defect recognition, small numbers of training epochs and batch sizes are generally adopted to ensure the fitting and convergence effectiveness of model training (Liu et al., 2022a, 2022b, 2023; Zeng et al., 2021). In order to avoid overfitting of DGRNN under the training of CDLM, the optimization strategies are carefully designed. In the pre-training stage of BK-Net, the number of training epochs is set to 20, the size of the input fabric image is 224 × 224, and the batch size of the input data is set to 32. At the same time, the Adam optimizer is used, the initial learning rate is set to 1e-6, the weight decay is set to 1e-4, and the cosine annealing update strategy is used to optimize the learning rate during model training. Since the size of D_m^{train} is twice that of the $D_m^{pre-train}$, considering the fitting efficiency and stability of the two-stage training of DGRNN, the selection of hyperparameters for the two-stage training is different. In the causality-guided training phase of DD-Net, the number of training epochs is set to 15, the input fabric image size is 224 × 224, and the batch size of input data is 32. The training also uses the Adam optimizer, the initial learning rate is set to 5e-4, the weight decay is set to 1e-4, and

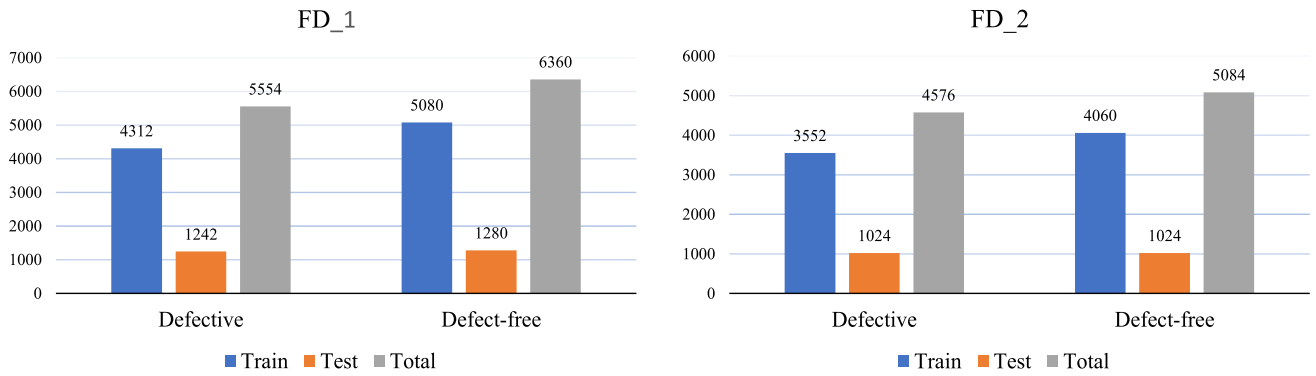


Fig. 8 The statistics of D_m^{train}

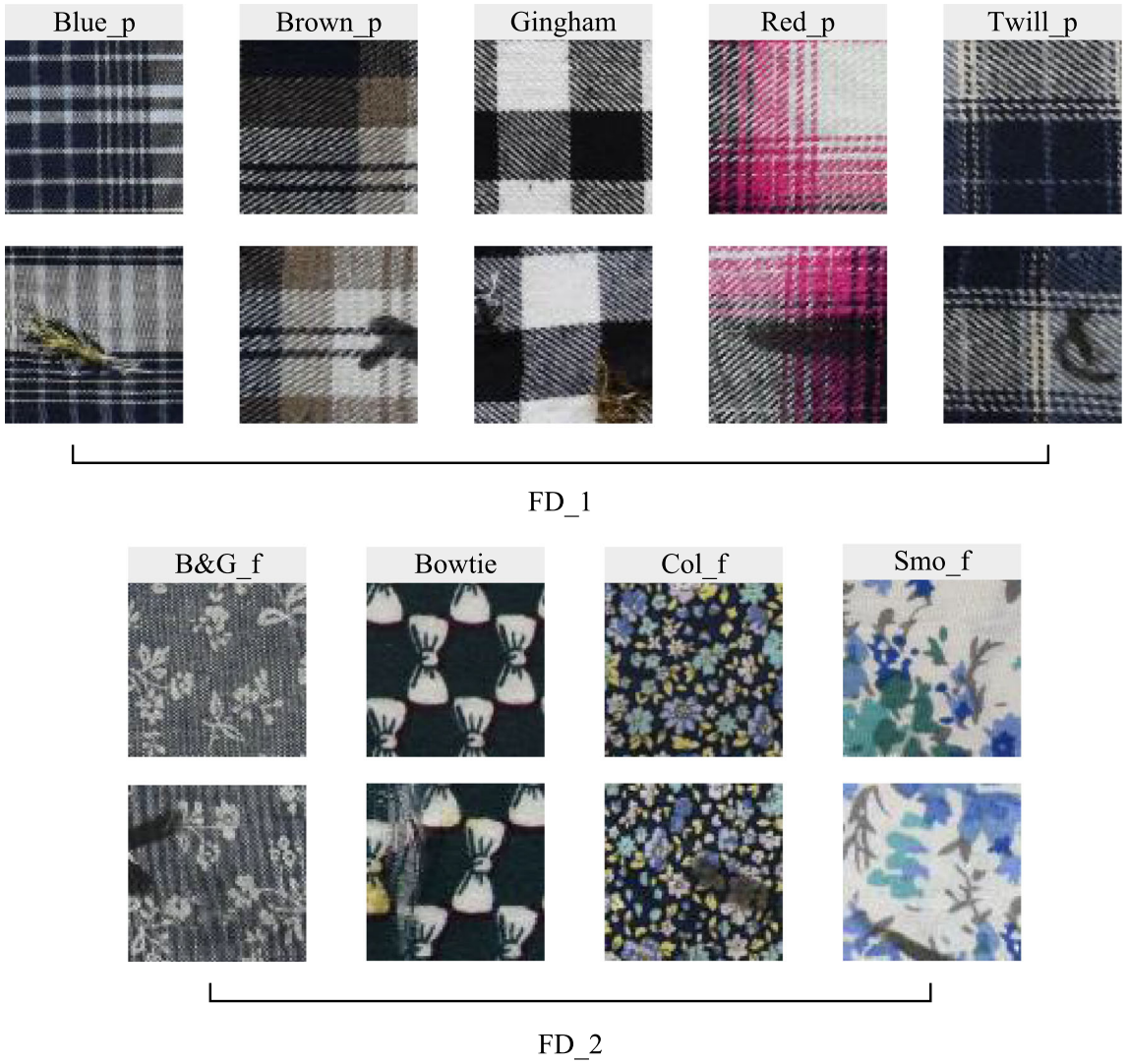


Fig. 9 The samples of D_m^{train}

the cosine annealing update strategy is used to optimize the learning rate during the model training process. At the same time, the input image will undergo two data preprocessing operations of random rotation and channel normalization. Besides, a comparison of different learning rate optimization strategies is presented in "Appendix".

Result and analysis

The experiments in this paper are divided into two parts, namely defect recognition performance evaluation and explainable feature recognition evaluation.

Defect recognition performance evaluation

The defect recognition performance evaluation consists of three parts, namely comparison study, parameter optimization study and ablation study. First, three mainstream convolutional neural networks and a fabric defect inspection method are selected for comparative study. Secondly, parameter optimization experiments are carried out for the selection of the causal expansion factor λ_i. Besides, ablation study is performed for the setting of CIM.

The indicators for defect recognition performance evaluation include two parts. The first part is used to evaluate the defect recognition and classification performance of the model, including ACC, PRE, REC, F1-score, and AUC. The second part is used to evaluate the recognition efficiency of the model in actual inspection scene, including PAR, FLOPs and FPS. ACC, PRE, REC and F1-score are used to measure the classification ability of the model for positive and negative samples respectively. Besides, AUC value represents the area under the model classification ROC curve, which can intuitively represent the classification performance of the model. PAR and FLOPs are used to evaluate how many parameters and floating-point operations a model contains, which represents the model complexity and the amount of calculation required for decision-making. FPS indicates how many frames or images the model can process per second, which can directly present the real-time inspection ability of the model.

Comparison study

In the comparison study, ResNet-50 (He et al., 2016), DenseNet-121 (Huang et al., 2017), SqueezeNet1_0 (Iandola et al., 2016) and Mobile-Unet (Jing et al., 2022) are selected, where the former three are the backbones commonly used for fabric defect recognition and Mobile-Unet performs feature extraction through the backbone MobileNetV2 (Sandler et al., 2018) and transfers the parameters to a segmentation network to achieve fabric defect inspection. Due to the role of the residual block, ResNet-50 can prevent key defect features from being overwhelmed and better capture the tiny features of the fabric image. DenseNet-121 uses a dense connection mechanism to enhance the ability to capture tiny image features. SqueezeNet is a lightweight network that has advantages in processing speed of fabric images. The comparison results of the training stage are shown in Fig. 10.

The training accuracy comparison of each model on the FD_1 training set is shown in Fig. 10a. It can be seen that all models tend to converge after 1500 iterations, and the final convergence accuracy rates of ResNet-50, DenseNet-121, SqueezeNet1_0 and Mobile-Unet are 90.79%, 88.82%, 71.89% and 87.22% respectively. The training accuracy comparison of each model on the FD_2 training set is shown in Fig. 10b. It can be seen that all models tend to converge after 1100 iterations, and the final convergence accuracy rates of ResNet-50, DenseNet-121, SqueezeNet1_0 and Mobile-Unet are 89.31%, 91.29%, 67.59% and 90.17% respectively. However, the accuracy of the proposed DGRNN reaches the highest 94.64% and 94.49% on FD_1 and FD_2 respectively when it converges, which preliminarily illustrates the excellent performance of the method proposed in this paper.

At the same time, all models are saved and further evaluated on the test set. The results of various indicators are shown in Tables 4, 5. From the perspective of the Acc index, the scores of all models have only slightly decreased, which indicates that none of the models have been over-fitted during the training phase. In addition, DGRNN reached 94.09%, 94.09%, 94.09%, 94.09% and 98.17% in ACC, PRE, REC, F1-score and AUC scores respectively on FD_1. Besides, the indicators of DGRNN on FD_2 also got the highest scores among all models, which shows the excellent ability of comprehensive classification of the model proposed in this paper. The rest of the models that perform better in the comparison study on FD_1 is ResNet-50, DenseNet-121, Mobile-Unet and SqueezeNet1_0. And the rest of the models that perform better in the comparison study on FD_2 is DenseNet-121, ResNet-50, Mobile-Unet and SqueezeNet1_0. These experimental results will be analyzed in detail below.

The existence of CIM contributes to the improvements of the recognition results of DGRNN in outperforming those comparison models. Under the training strategy of CDLM, DGRNN can capture the enhanced differences between defects and patterns through causal intervention in each iteration to strengthen the explainable learning of defect features. This causality-guided feature learning method improves detection accuracy on the basis of enhancing the explainability of the model. Besides, there are mainly two reasons for the weak performance of the comparison models, namely the unexplainability of visual feature sampling and the inter-layer residual linking mechanism of some DNN. A detailed analysis based on these two points of comparison models are conducted as follows.

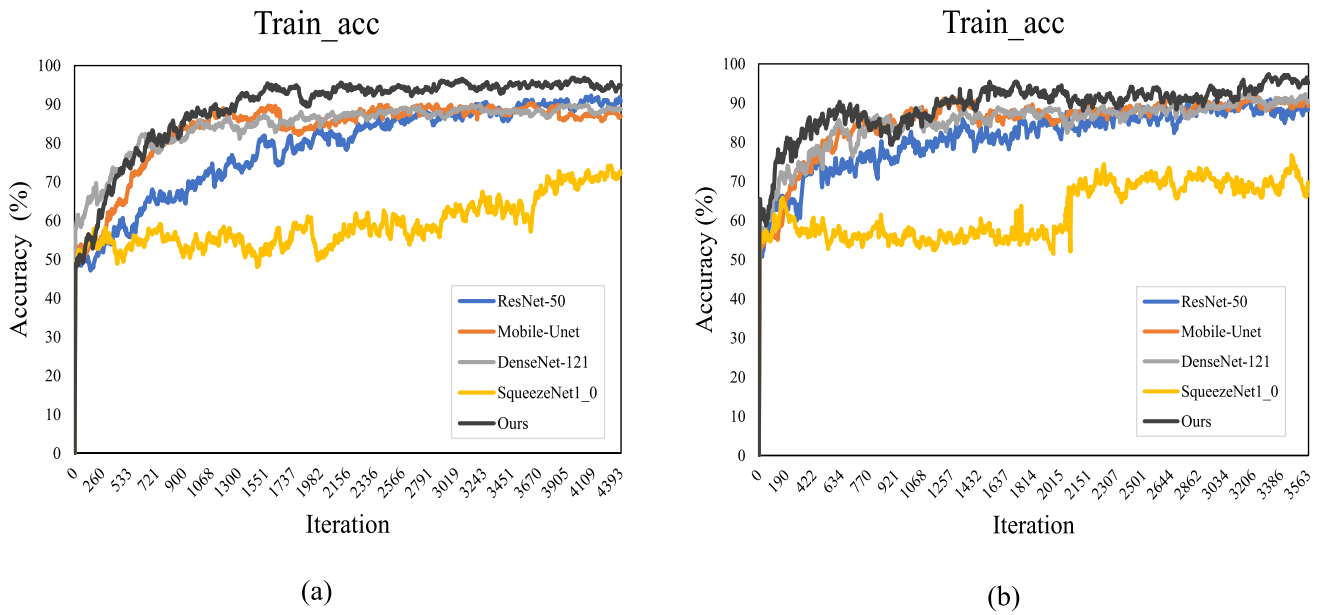


Fig. 10 The accuracy comparison during training

Table 4 The performance comparison with cutting-edge methods on FD_1

Metrics	Models				
	ResNet-50	DenseNet-121	SqueezeNet1_0	Mobile-Unet	Ours (λ_{1234} = 0.01)
ACC (%)	88.86	86.28	70.78	86.16	94.09
PRE (%)	89.01	86.50	74.22	86.29	94.09
REC (%)	88.71	86.34	71.05	86.21	94.09
F1-score (%)	88.83	86.27	69.87	86.16	94.09
AUC (%)	95.65	92.31	74.49	91.35	98.17

Table 5 The performance comparison with cutting-edge methods on FD_2

Metrics	Models				
	ResNet-50	DenseNet-121	SqueezeNet1_0	Mobile-Unet	Ours (λ_{1234} = 0.01)
ACC (%)	88.96	90.43	65.67	87.60	93.95
PRE (%)	89.71	90.45	71.74	87.62	94.20
REC (%)	88.96	90.43	65.67	87.60	93.95
F1-score (%)	88.91	90.43	63.10	87.60	93.94
AUC (%)	96.04	96.72	81.22	95.24	98.22

First, the poor performance of these models is essentially limited by the poorly explainable process of sampling visual features by DNN. Without causal knowledge and intervention, DNN is unable to truly recognize the defects and distinguish the difference between defects and background patterns under visual interference, and then fall into pixel-wise sample selection bias and false correlations obstacle.

Second, the existence of the residual linking mechanism, e.g., ResNet-50, DenseNet-121 and Mobile-Unet, can also enhance the unexplainability of DNN, resulting in poor recognition performance. In a linking unit, residual operation will fuse the original input and output, which leads to secondary interference of pattern information, and seriously

Table 6 The recognition efficiency comparison with cutting-edge methods

Metrics	Models				
	ResNet-50	DenseNet-121	SqueezeNet1_0	Mobile-Unet	Ours ($\lambda_{1234} = 0.01$)
PAR (M)	23.51	6.956	0.736	2.226	4.972
FLOPs (G)	4.13	2.90	0.73	0.33	0.87
FPS	11.12	11.24	43.40	26.69	35.57

Table 7 The causal expansion factor optimization results on FD_1

λ	Acc (%)	Pre (%)	Rec (%)	F1-score (%)	AUC (%)
$\lambda_{1,2,3,4} = [0.01,0.01,0.01,0.01]$	**94.09**	**94.09**	**94.09**	**94.09**	**98.17**
$\lambda_{1,2,3,4} = [0.01,0.02,0.03,0.04]$	92.86	92.86	92.87	92.86	97.66
$\lambda_{1,2,3,4} = [0.04,0.03,0.02,0.01]$	90.05	90.11	90.02	90.04	96.55

Bold indicates superiority of the model with the proper parameter combination

affects the model to capture defect features. In fact, networks with residual linking mechanisms have experienced performance degradation on some other industrial inspection problems (Cheng et al., 2022; Wang et al., 2021). Besides, differences in linking mechanisms can also affect model performance. ResNet-50 and Mobile-Unet are models with local linking mechanisms, i.e., each residual block is combined in series. Under the local linking mechanism, the pattern interference is limited to the sampling stage at a single scale, and the transmission of interference is small. This also explains the similar performance gap between the two models on FD_1 and FD_2 datasets. However, DenseNet-121 is a model with global linking mechanism and the interference is with strong transmission, which means its performance is greatly affected by the patterns and their types. The difference of background pattern of CPF in FD_1 and FD_2 results in the difference of feature maps in different scales. The feature map of FD_1 retains similar and regular pattern information in multiple scales, and situation in FD_2 is the opposite. This makes it more difficult for DenseNet-121 to capture the difference between defects and patterns on FD_1 under the global linking mechanism, resulting in poor recognition performance.

Compared with all other models, although SqueezeNet1_0 is the most lightweight, its performance is poor because it hardly considers the connection between multi-layer features. However, CPF images usually contain rich and relatively chaotic semantic information, especially on FD_2. At the same time, using the maximum pooling layer for down-sampling causes a large amount of loss of semantic information resulting in the bad performance.

In order to ensure that DGRNN can be further implemented to industrial-level defect inspection scenes, the recognition efficiency comparison experiments are performed and the results is shown in Table 6. In terms of PAR indicators, the model parameters of the model proposed in this paper are the third to SqueezeNet1_0 of 0.736 million and Mobile-Unet of 2.226 million, and are reduced by 1.4 times and 4.8 times compared to DesenNet-121 and ResNet-50 respectively. Besides, the FLOPs of proposed model are also the third among all models. This indicates that the proposed model has the average complexity. However, the FPS of proposed model reaches the 35.57 and is only less than the SqeeuzeNet1_0 of 43.40. Based on the previous studies of the timeliness of the fabric defect inspection (Wang et al., 2022a, 2022b; Zhao et al., 2020a, 2020b), the decent FPS that a real-time inspection model should reach around 30, which means the DGRNN can meet the requirements of actual inspection.

Parameter optimization study

In order to explore the effectiveness of the feature difference weight sets of various scales on the performance of the model, three sets of parameter optimization experiments of the causal expansion factor λ_i are carried out.

The parameter settings are the average parameter combination, the forward arithmetic parameter combination and the reverse arithmetic parameter combination respectively. The results of each group of parameter optimization experiments on two datasets are shown in Tables 7 and 8.

On FD_1, It can be seen from the comparison that the forward arithmetic parameter combination can assist the model achieve better performance than the reverse arithmetic parameter combination. This shows that for the CPF data on FD_1, the weight of the deeper scale is more important for the defect recognition performance. This is because the background patterns of the CPF images on FD_1 are regular geometric lines. At the same time, the difference in geometric semantic information between different CPF is small, resulting in richer information contained in the deeper-scale features after multi-layer information fusion. In addition,

Table 8 The causal expansion factor optimization results on FD_2

λ	Acc (%)	Pre (%)	Rec (%)	F1-score (%)	AUC (%)
$\lambda_{1,2,3,4} = [0.01,0.01,0.01,0.01]$	**93.95**	**94.20**	**93.95**	**93.94**	**98.22**
$\lambda_{1,2,3,4} = [0.01,0.02,0.03,0.04]$	90.67	90.68	90.67	90.67	97.04
$\lambda_{1,2,3,4} = [0.04,0.03,0.02,0.01]$	91.41	91.41	91.41	91.41	97.71

Bold indicates superiority of the model with the proper parameter combination

Table 9 The ablation results on FD_1

CIM	Acc (%)	Pre (%)	Rec (%)	F1-score (%)	AUC (%)
$FDM_{1,2,3,4} = [\times, \times, \times, \times]$	87.59	88.14	87.68	87.56	96.84
$FDM_{1,2,3,4} = [\surd, \times, \times, \times]$	90.09	90.30	90.03	90.06	95.90
$FDM_{1,2,3,4} = [\times, \surd, \times, \times]$	89.77	90.68	89.66	89.69	95.65
$FDM_{1,2,3,4} = [\times, \times, \surd, \times]$	88.14	90.12	87.98	87.96	95.13
$FDM_{1,2,3,4} = [\times, \times, \times, \surd]$	91.36	92.67	91.23	91.27	97.35
$FDM_{1,2,3,4} = [\surd, \surd, \times, \times]$	90.17	91.40	90.04	90.07	95.88
$FDM_{1,2,3,4} = [\surd, \surd, \surd, \times]$	90.96	90.96	90.96	90.96	97.18
$FDM_{1,2,3,4} = [\surd, \surd, \surd, \surd]$	**94.09**	**94.09**	**94.09**	**94.09**	**98.17**

Bold indicates superiority of the model with the proper parameter combination
× and √ denotes the absence or presence of FDM on this scale

Table 10 The ablation results on FD_2

CIM	Acc (%)	Pre (%)	Rec (%)	F1-score (%)	AUC (%)
$FDM_{1,2,3,4} = [\times, \times, \times, \times]$	89.84	89.93	89.84	89.84	97.34
$FDM_{1,2,3,4} = [\surd, \times, \times, \times]$	91.75	92.77	91.75	91.70	98.09
$FDM_{1,2,3,4} = [\times, \surd, \times, \times]$	91.46	92.17	91.46	91.42	96.41
$FDM_{1,2,3,4} = [\times, \times, \surd, \times]$	90.23	90.72	90.23	90.21	95.75
$FDM_{1,2,3,4} = [\times, \times, \times, \surd]$	90.91	90.97	90.92	90.92	97.00
$FDM_{1,2,3,4} = [\surd, \surd, \times, \times]$	91.94	91.96	91.94	91.94	97.31
$FDM_{1,2,3,4} = [\surd, \surd, \surd, \times]$	93.21	93.23	93.21	93.21	**98.35**
$FDM_{1,2,3,4} = [\surd, \surd, \surd, \surd]$	**93.95**	**94.20**	**93.95**	**93.94**	98.22

Bold indicates superiority of the model with the proper parameter combination

the performance of the model is the best under the average parameter combination, which shows that comprehensively considering the difference weight of each scale is more helpful for the model to make correct decisions.

On the contrary, the reverse arithmetic parameter combination can make the model perform better than the forward arithmetic parameter on FD_2. This shows that for the CPF data on FD_2, the difference weight of the shallower scale is more important to the defect recognition performance. This is because the geometric semantic information of the CPF image of FD_2 is irregular curves, and the distribution is relatively complex, which means that the shallow-scale features contain richer semantic information and has more important impact on the recognition performance of the model. However, the model performance under the average parameter combination is also the best on FD_2, which means that on the FD_2, it is also more helpful for model recognition to comprehensively consider the features of each scale.

Ablation study

After determining the use of the average parameter combination, in order to explore the influence of the multi-scale FDM combination in CIM on the performance of the DGRNN, eight sets of ablation experiments are carried out and the results on two datasets are shown in Tables 9 and 10 respectively.

The CIM proposed in this paper is actually a parallel combination of four FDM and the ablation study includes three ablation settings. The first ablation setting is DGRNN without CIM, which means no causal intervention is performed at the four scales. The second ablation setting contains four experiments, where CIM performs causal intervention on

four scales. The third set of ablation settings consists of three experiments, where multiple causal interventions are simultaneously executed at different scales. Three conclusions can be drawn from the comparison on FD_1. First, the existence of CIM effectively improves the performance of the model. Second, the deep-scale FDM in CIM has the greatest impact on the performance of the model, which shows that implementing causal intervention on the deep-scale features is more effective. Third, the comprehensive causal intervention on the features extracted from four scales has the best effect on improving the model performance, which illustrates the effectiveness of multi-scale causal intervention strategy. The effectiveness of multi-scale CIM can also be seen from the results on FD_2. However, unlike FD_1, causal intervention on shallow-scale features works better on FD_2. This is also caused by the difference in semantic information between CPF with stripe and CPF with irregular geometric curve.

Explainable feature recognition evaluation

The explainable feature recognition evaluation consists of two parts, namely the defect down-sampling study and the explainable decision-making study using Grad-CAM (Selvaraju et al., 2017).

Defect feature down-sampling study

As shown in Fig. 11, in order to analyze the process of defect extraction of DGRNN, the feature maps extracted from four scales are visualized.

According to the results, it can be seen that DGRNN can roughly locate defects on the shallower scale. However, it does not completely ignore background features, because the feature maps output from the previous layers are still large and contain rich semantic information. On the deeper scale, DGRNN gradually concentrates on sampling the area of defect features, which preliminarily illustrates the effectiveness of the proposed method for resisting the background interference.

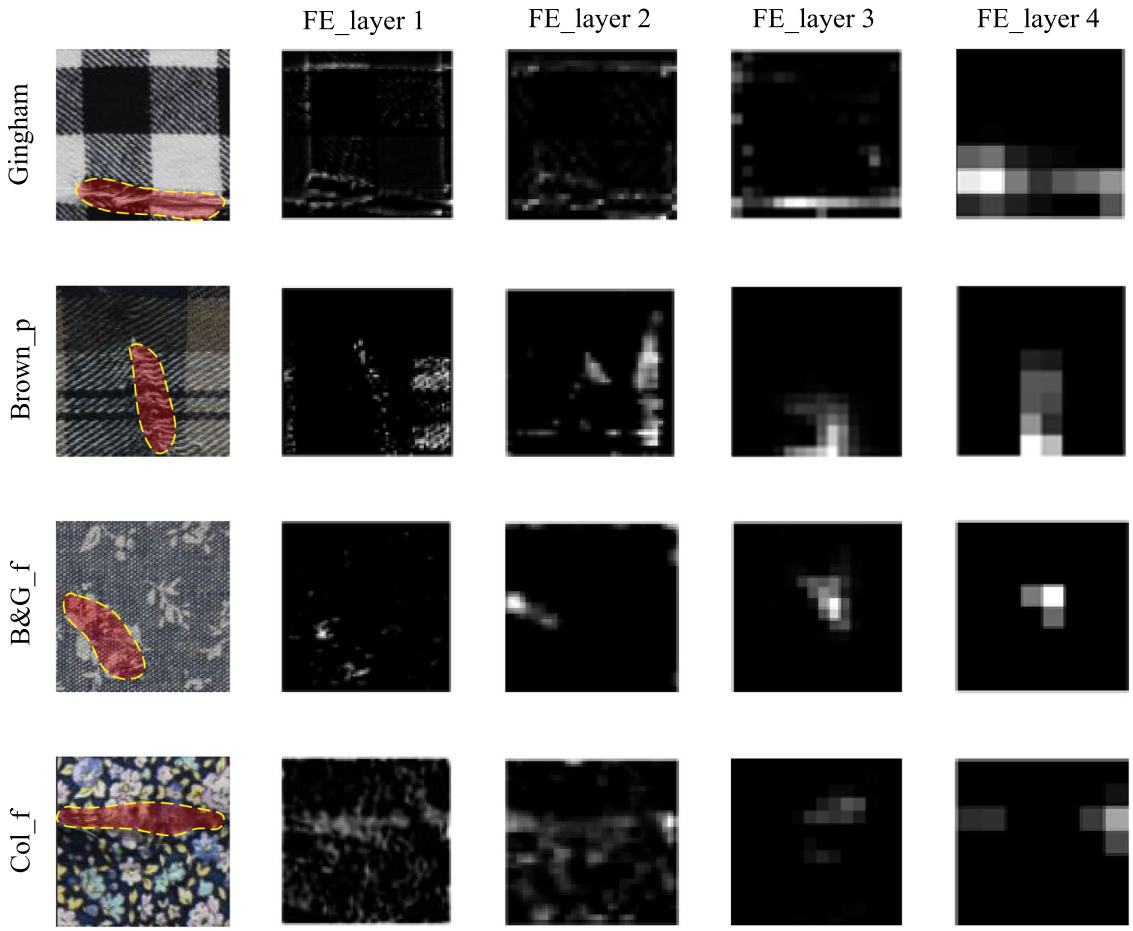


Fig. 11 The feature maps from different scales of DGRNN

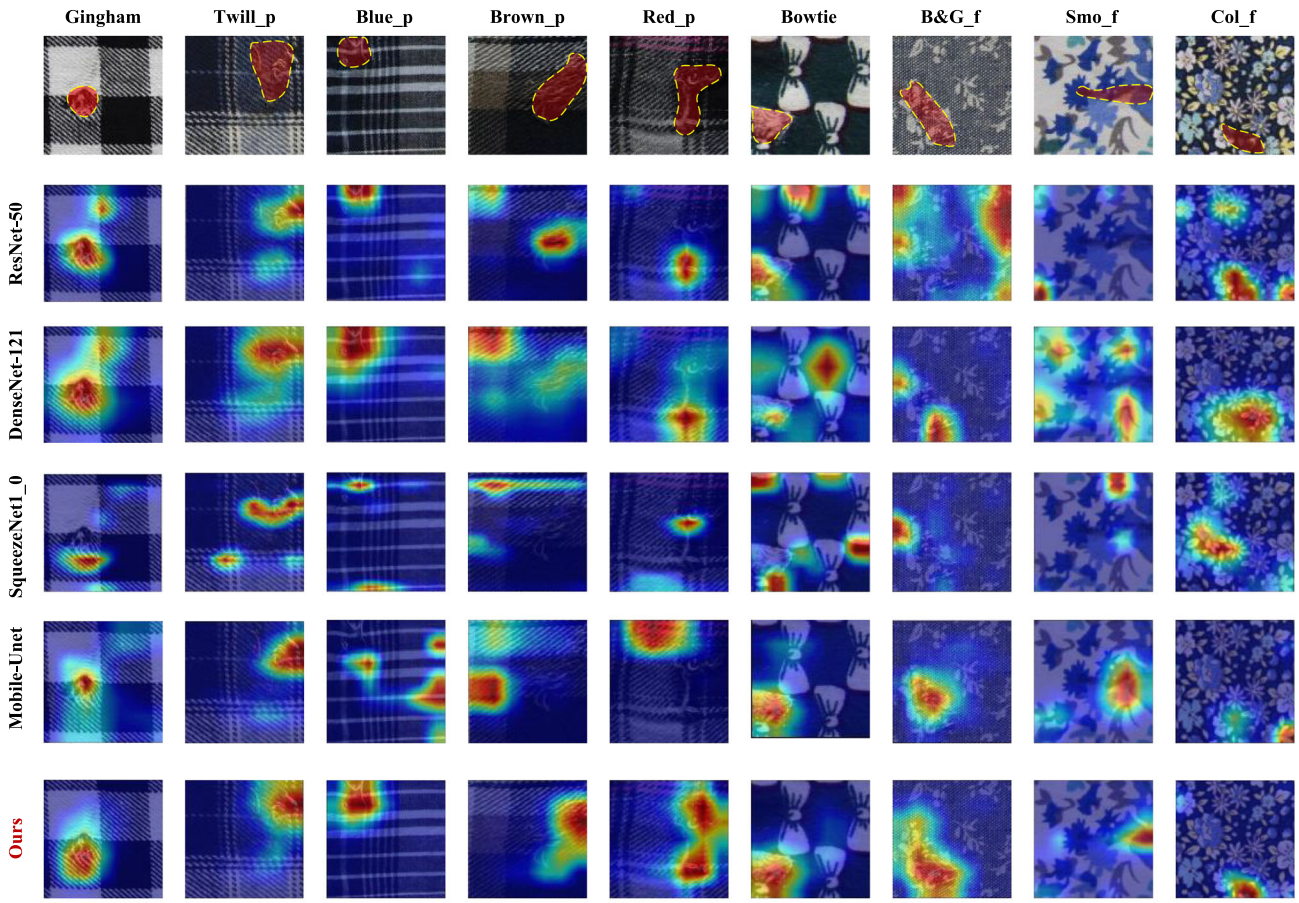


Fig. 12 The comparison of Grad-CAM results of different cutting-edge methods

Explainable decision-making study

The explainable decision-making study is described secondly. In order to demonstrate that the running process of DGRNN is explainable, the Grad-CAM method is used to compare the decision-making of each model. Grad-CAM is a visualization technique that assists to analyze which area of an image that DNN samples to make a decision on its class and represented as the brightest area in the image. As shown in Fig. 12, ResNet-50, DenseNet-121, SqueezeNet1_0, and Mobile-Unet all fall into a spurious correlation situation. For example, on a Bowtie image, ResNet-50 pays attention to the position of the defect, but at the same time diverges attention to the background pattern which is similar to the defect feature. DenseNet-121 also diverts a lot of attention to background patterns when recognition the defect of Smo_f images. In contrast, the proposed method can ensure the accurate and explainable recognition of defects while successfully avoiding the interference of background patterns. This shows that DGRNN realizes the decision on the defective fabric label by locating and sampling the features of defect area, which further proves that its decision-making process follows the learned explainable causality.

In addition, DGRNN is visualized at epoch 3, 6, 9, 11, 13, and 15, and the results are shown in Fig. 13. It can be seen from that DGRNN can gradually avoid the interference of background patterns after epoch 6, and accurately extract the area of the defect to establish the explainable data correlation. During the iteration, DGRNN can fully focus on the defect features. Taking Gingham fabric as an example, in the initial iteration stage, e.g., epoch 3, although DGRNN paid attention to the abnormality of the defect position, it still focused on some background patterns. It shows that at this stage, DGRNN cannot really identify defects based on the characteristics of defects. But in the epoch 6 and epoch 9 stages, it can be clearly seen that DGRNN is constantly focusing on the abnormal defect area. In the epoch 11 stage, DGRNN has completely focused on the defect area, indicating that at this time DGRNN can already establish the relationship between defects and labels, and truly identify defects. This shows that the proposed CDLM guides the DGRNN to optimize on the explainable learning direction of causality, and

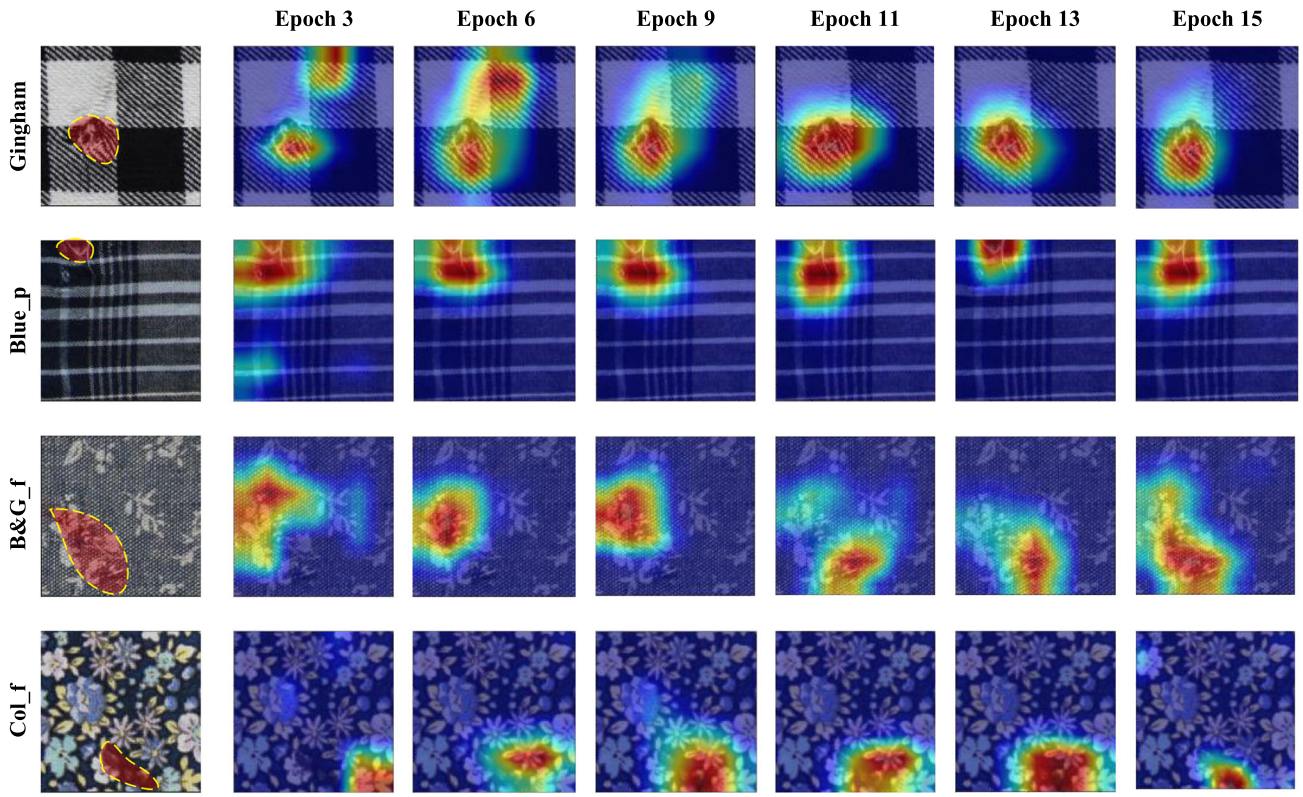


Fig. 13 The comparison of Grad-CAM results of DGRNN at different training stages

successfully capture the data correlation between defect features and defective fabric labels under visual interference.

Conclusion

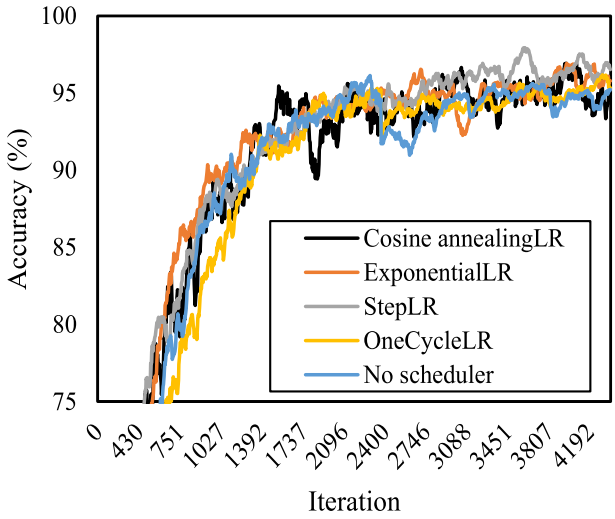
To overcome the visual interference of background to achieve accurate defect recognition of CIP, this paper first introduces the causal inference into the explainable vision-based quality inspection and proposes a CDLM that enables data-driven inspection model make explainable decisions. First, a SCM for the defect recognition of CIP is constructed and a causal intervention strategy for blocking background interference is obtained. Second, feature-wise causal intervention and accurate and explainable learning of defect features are achieved using DGRNN with a multi-scale feature difference mechanism. The main contributions of this paper are as follows:

(1) To the best of our knowledge, this is the first attempt by integrating causal inference into DL-based quality inspection of CIP. The results indicating that building a causal deep learning model is effective to concentrate the detector on the defective areas of CIP. As a result, the detector is not sensitive with the background of input images, which provides a novel way to realize artificial general intelligence from the perspective of practical investigation in computer vision.
(2) A CDLM is proposed for defect recognition of CIP, which consists of two parts. First, a DGRNN with a combination of multiple sub-networks and CIM with feature difference mechanism is proposed to tackle the graphic inherency interference and graphic similarity interference in defect recognition of CIP. Second, a combined intervention strategy of multi-scale FDM and causality expansion factors is proposed to tackle the graphic difference interference in defect recognition of CIP.

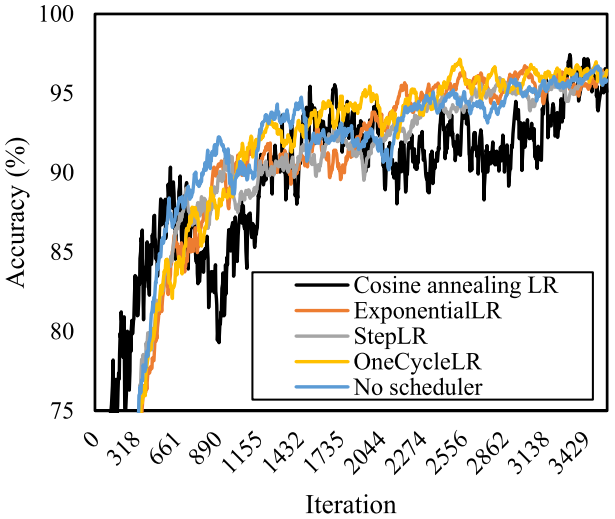
Future works will be conducted in two parts. First, this method does not solve the problem of recognition of multiple types of defects in CIP. Second, hardware and software have not further been designed to integrate with this method and been applied to the actual inspection scenarios.

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Data availability The experimental data used in this study are from 'ZJU-Leaper: A Benchmark Dataset for Fabric Defect Detection and a



(a) Comparison of various learning rate strategies on FD_1



(b) Comparison of various learning rate strategies on FD_2

Fig. 14 Comparison of different learning rate optimization strategies

Comparative Study' (Zhang et al., 2020a, 2020b). The DOI of this article is https://doi.org/https://doi.org/10.1109/tai.2021.3057027 and the data are available by visiting https://github.com/nico-zck/ZJU-Leaper-Dataset or http://www.qaas.zju.edu.cn/zju-leaper/.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

The setting of hyperparameters in the training of DNN is always an empirical problem. In this paper, five different learning rate optimization strategies are selected for experiments, namely constant learning rate, i.e., no scheduler, cosine annealing learning rate, exponential learning rate and its parameter gamma is set to 0.98, step learning rate and its parameter step is set to 5 and gamma is set to 0.9, and one cycle learning rate. The comparison results are shown in Fig. 14.

The results show that different learning rate optimization strategies have slight differences in the training of CDLM, but all of these strategies can contribute to a decent score. In this paper, the cosine annealing optimization strategy is chosen because it can make the DGRNN achieve faster convergence speed and more efficient fitting result. Besides, as shown in the training results on the FD_2 data set, it can be seen that the cosine annealing strategy can enable DGRNN to achieve a higher recognition accuracy in the early iterative training stage. However, it also has some shortcomings, such as the existence of certain oscillations after convergence. Therefore, trying different learning rate optimization strategies to assist the training of DGRNN in other visual inspection scenarios is encouraged.

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